

Ibrahim Haykal Alatas

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SUMMARY

Backend/System Engineer with experience building internal systems on top of legacy ERPs (Infor/Baan) to support manufacturing operations across warehouse, production, and accounting workflows. Focused on transforming raw operational data into reliable digital systems that handle real-world constraints like strict bureaucracy, transactional consistency, and cross-department tracking.

EXPERIENCE

Full Stack Developer (System Engineer) Intern

Feb. 2025 – Present

PT Gemala Kempa Daya (Astra Otoparts Group)

Jakarta, Indonesia

- **Raw Material Warehouse Inventory:** Engineered a real-time inventory system with FIFO logic and dynamic rack mapping (48 blocks) supporting 400+ weekly transactions by bypassing read-only ERP constraints.
- **Trucking Control System:** Executed phased database migration from Oracle to PostgreSQL utilizing an application-level dual-write strategy to maintain zero downtime during live production.
- **Smart Andon System:** Designed a stateful ticketing architecture mapping the physical Andon workflow, measuring Mean Time To Repair (MTTR), and implementing OTP/QR validation.
- **Accounting & Cost Management:** Developed financial reporting modules for inventory movement, optimizing heavy Excel computations using client-side rendering (SheetJS) to reduce server memory load by 50%.

Web Developer

Aug. 2024 – Jan. 2025

Nirmala Technology

Jakarta, Indonesia

- Designed and engineered a bilingual corporate profile platform utilizing Laravel 11 and Tailwind CSS.
- Developed UI components for the Q-Tin Dashboard utilizing DaisyUI and Flowbite ensuring responsiveness.

PROJECTS & ACHIEVEMENTS

SME Digital Platform (Hackathon 2nd Place) | *Laravel, Tailwind, WA API*

Jun. 2025

- Awarded 2nd Place nationally by developing a platform for MSMEs, integrating a WhatsApp chatbot to streamline user engagement and digital marketing.

AI Defect Detection System | *YOLOv8, Roboflow, Streamlit, WebRTC*

Dec. 2024

- Engineered an end-to-end visual inspection pipeline for automotive parts, performing manual dataset annotation and model training via Roboflow.
- Deployed a live real-time inference web application using Streamlit and WebRTC to demonstrate automated Quality Control workflows.

SIPS (Android Archiving System) | *Kotlin, Laravel 10, UML*

Sep. 2024 – Dec. 2024

- Conducted a comparative study at BBSPJIKFK (Ministry of Industry) to benchmark archiving workflows and architected the system via comprehensive UML diagrams and database structures.
- Executed full-stack development, delivering a robust REST API and a mobile user interface.

EDUCATION

Politeknik STMI Jakarta (Ministry of Industry)

Jakarta, Indonesia

Applied Bachelor in Computer Science - Information Systems; GPA: 3.77/4.00

Sep. 2022 – Present

- **Coursework:** Database Management (SQL), System Analysis & Design, Web Engineering, Data Structures.
- **Certifications:** Junior Web Developer (BNSP - 2023), Programming Foundations (Dicoding - 2023).

TECHNICAL SKILLS

Core Domain: Enterprise Architecture, Transactional System Design, ERP Data Processing (Infor/Baan), Business Process Modeling (BPMN), Data Transformation & Validation

Backend & Database: Laravel, PostgreSQL (Optimization, Migration, Dual-Write), Oracle PL/SQL, RESTful APIs

Tools & Practices: Git, Roboflow, WebRTC, JasperReports, SheetJS, Figma, Clean Architecture Principles